DRAWING MACHINE

The Turtle controller

The first part of the program is a function that moves the turtle, one command at a time. It is planned out in the flowchart on the previous page. This code enables the turtle to convert the "do" and "val" values into movement commands.

2 This code creates the Turtle controller function. It turns "do" inputs into directions for the turtle, and "val" inputs into angles and distances.

```
from turtle import *
def turtle_controller(do, val):
    do = do.upper()
    if do == 'F':
        forward(val)
    elif do == 'B':
        backward(val)
    elif do == 'R':
        right(val)
    elif do == 'L':
        left(val)
    elif do == 'U':
        penup()
    elif do == 'D':
        pendown()
    elif do == 'N':
        reset()
    else:
        print('Unrecognized command')
```

Loads all the commands that control the turtle

Defines "do" and "val" as inputs for the function

This command converts all the letters in "do" to upper case (capital letters)

This tells the function to turn a "do" value of F into the turtle command "forward"

As in the flowchart, the function checks the "do" letter against all the letters it understands

This command tells the turtle to start drawing on the page

This command instructs the turtle to stop drawing on the page

This command resets the turtle's position to the center of the screen

This message appears if the "do" value is a letter that the function cannot recognize

The starting position of the turtle

3 Here are some examples of how to use the Turtle controller. Each time it is used, it takes a "do, val" command and turns it into code the turtle can understand.

```
>>> turtle_controller('F', 100)
>>> turtle_controller('R', 90)
>>> turtle_controller('F', 50)
```

This calls the function using its name

These "do" and "val" inputs tell the turtle to move 100 steps forward

This makes the turtle turn right 90 degrees